ASSIGNMENT # 2
BT102: MICROBIOLOGY
Spring 2021

Total Marks: 10
Due date: 19th July 2021

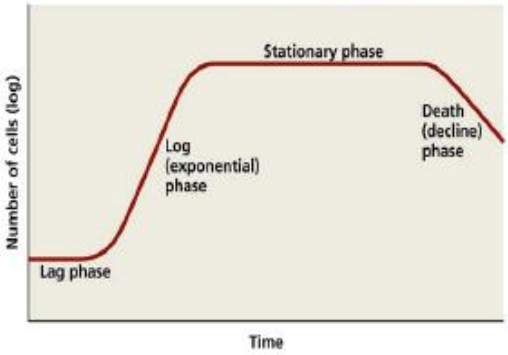
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Questions:

(A) Explain the bacterial growth curve and how can we estimate the microbial growth?

Answer:

Growth Curve

- Batch Culture
  - Liquid medium in a closed vessel
  - No fresh medium
- Nutrients depleted
- Waste products increased



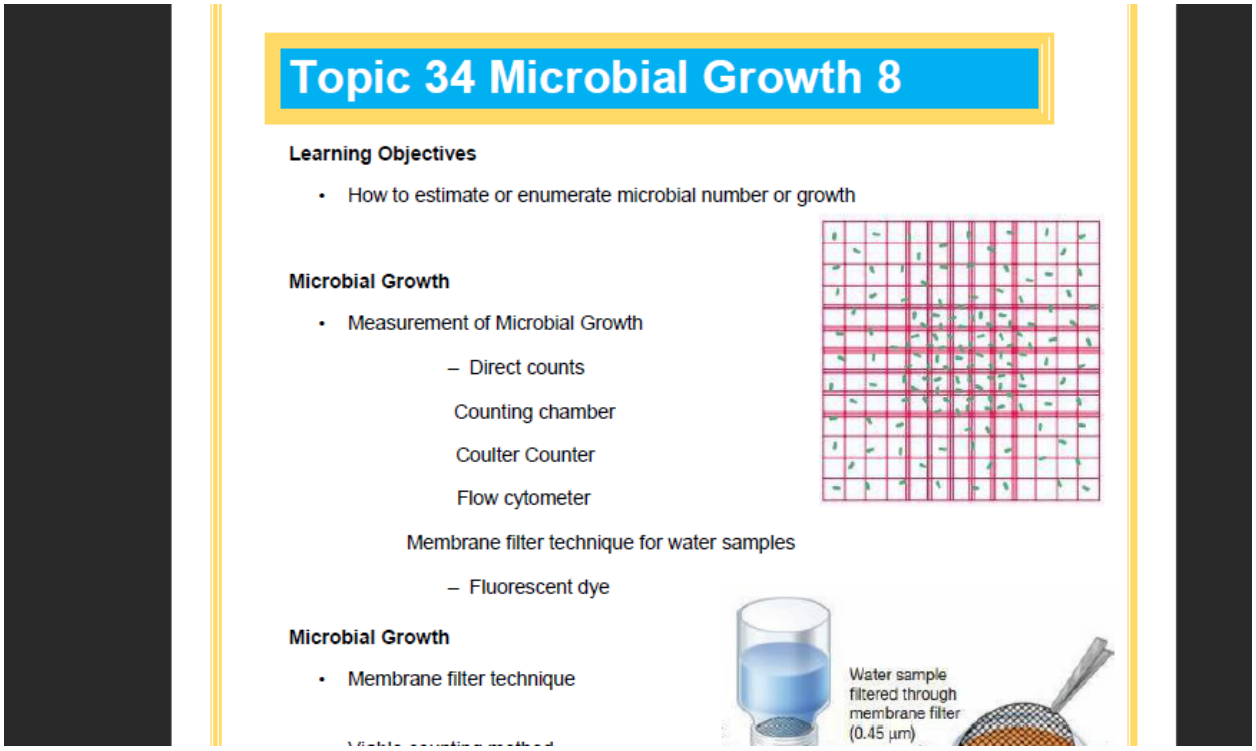
- Generation time
- Doubling time
- Can be calculated from the exponential phase of growth

$$k = \frac{\log 10^9 - \log 10^3}{(0.301)(10\ \text{hr})} = \frac{9-3}{3.01\ \text{hr}} = 2.0\ \text{generations/hr}$$

$$g = \frac{1}{2.0\ \text{gen. hr}} = 0.5\ \text{hr/gen. or } 30\ \text{min/gen.}$$

Uses of Growth Curve

- Growth dynamic of any bacteria can be studied
- Exponential phase is ensured for mass production of microbes



(B) What are the growth requirements of microbes, discuss briefly.

Answer:

Growth Requirements (For details read handouts topic #27-30)
Physical:
1. Temperature
2. pH
3. Osmotic Pressure

Chemical:
1. Source of C
2. N, S, P, O,
3. Trace elements
4. Growth factors

(C) A pastry chef accidentally inoculated a cream pie with six *S. aureus* cells. If *S. aureus* has a generation time of 60 minutes, how many cells would be in the cream pie after 7 hours?

ANSWER: (Google)

Number of cells x 2^(n) generations = total # of bacteria present after n generation

Generation time = 60 minutes: the time it takes an organism to divide into 2 cells; it is the time required for the population of cells to double.
In 1 hour 1 cell divides into 2

Therefore, number of cells after would be 2^(7) x 6
6 x 2^(7) = 768
Therefore, there will be 768 cells in the cream pie after 7 hours.

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